Cloud Computing Security: Attacks and solutions

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ABSTRACT

Cloud computing has been visualized as the future generation technology of enterprises. It not only makes the services cost effective but also improves productivity. The earlier traditional approaches where all the services are in actual control of the user, in contrast to which, cloud computing place all the database files and various system and application software's to data center, which make the system vulnerable to different security threats. Though, cloud computing has lot of advantages, but on the security front there still exists a lot of challenges which needs further consideration. In this paper, different types of attacks faced by clouds have been explained. Further, proposed solution for attacks has been discussed.

KEYWORDS

Cloud computing, Virtualization, Datacenters, Virtual machines

1. INTRODUCTION

In the field of technology, there have been different approaches for improving the parallelism and sharing of different resources for the development and facilitation of data usage. Various techniques has been introduced in the past like clustering, data base management system, grid system, and many others. Cloud computing is recently emerging as a technology for performing complex computations, and for acting as repository for storing the resources. It allows the users to customize their resource usage according to their needs. Cloud allows big enterprises to reduce their financial burden in terms of Capital and Operational Expenditure which give them upper hand in the competitive world. According to NIST, cloud computing[1,2] is a model for enabling convenient on-demand network access to a share pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort or service provider interaction. With the rapid development of processing and storage technologies, cloud computing has recently emerged as a new computing model for hosting and delivery services over the internet. By using this resources provide general utilities that can be leased and release by user through the interned in an on demand fashion. To make these model more useful services providers are used in this type of cloud involvement. Service provider is divided into two parts firstly the infrastructure provider who helped in managing the cloud and secondly the service provider who used to rent resources from one or many in fracture provider to serve the multiple end users[3,4]. The main reason for the existence of different perceptions of cloud computing is that cloud computing is not only a new technology but rather a new operation models that brings together a set of existing technologies to run business in different way. Cloud computing[5] is often compared to following technologies, some of them are:-

- Grid Computing-Grid computing[6] is a distributed architecture of large numbers of computers connected to solve a complex problem. It is a server that runs independent tasks and are loosely linked by the Internet or low-speed networks. Computers may connect directly or via scheduling systems.
- Utility Computing-Utility computing presents the model of providing resourced-on-demand and charging customers based on usage. Utility computing is a computing business model in which the provider owns, operates and manages the computing infrastructure and resources, and the subscribers accesses it as and when required on a rental or metered basis.
- Virtualization-It is a technology that abstracts away the details of physical hardware and provides virtualized resources for high level applications. It is the creation of a virtual rather than actual version of something, such as an operating system, a server, a storage device or network resources
- Autonomic computing- Autonomic computing is a self-managing computing model named after, and patterned on, the human body's autonomic nervous system. It would control the functioning of computer application. The goal of autonomic computing is to create systems that run themselves, capable of high-level functioning while keeping the system's complexity invisible to the user.

In contrast to traditional networks, cloud offer flexibility and robustness. In the last management layer protects the system from different rates of traffic with the help of adding and removing servers. This layer also has the capability to secure the cloud. In Cloud various services models are defined as below:

- INFRASTRUCTURE AS A SERVICE (IAAS)
  IAAS refers to on-demanded provisioning of infrastructure resources like Processor, memory, networks, and other resources. It allows users to use different software's like operating systems, and any other application software's. Usually in terms of UMS the cloud owners who officers IAAS is called IAAS provider exp-Amazon EC2.
- PLATFORM AS A SERVICE (PAAS)
  PaaS delivers the user ability to post onto the cloud any of the consumer developed or acquired application using programming Language and tools. In this service user does not manage any of the cloud infrastructure like servers, storage, Operating System, but can manage the deployed application.
- SOFTWARE AS A SERVICE (SAAS)
  SAAS refers to providing on-demand application over the internet. User can access any application which is running on a cloud infrastructure. These applications can be accessed through client server interface like web browser. The user cannot manage the cloud infrasture in this model. For example: Rockspace.

It is understood that cloud computing is the advancement in the world of on demand information technology services and products. Various other traditional approaches which benefitted the IT services are AEC08, IBM07c and VCL04 [3] etc. The term which got its name from the cloud symbol used in basic networking figure and became popular in 2007 after the joint collaboration of IBM and google. Though, cloud offers lot of benefits above other techniques like scalability, faster implementation, higher utilization of resources, etc., but technology suffers from lot of challenges. Some of the most important are as follows:

- SECURITY AND PRIVACY
  The main challenge to cloud computing is how to address the security and privacy concerns of businesses thinking of adopting it. The fact that the valuable enterprise data will reside outside the corporate firewall raises serious concerns. Hacking and various attacks to cloud infrastructure would affect multiple clients even if only site is attacked. These risks can be mitigated by using security applications, encrypted file systems, data loss software, and buying security hardware to track unusual behavior across servers. It is difficult to assess the costs involved due to the on-demand nature of the services. Budgeting and assessment of the cost will be very difficult unless the provider has some good and comparable benchmarks to offer. The service-level agreements of the provider are not adequate to guarantee the availability and scalability. Businesses will be reluctant to switch to cloud without a strong quality guarantee.
- INTERPERABILITY AND PORTABILITY
  Business should have the leverage of migrating in and out of the cloud and switching providers whenever they want, and there should be no lock-in period. Cloud computing services should have the capability to integrate smoothly with the on-premise IT. Since many companies are still not sure about its implementation, the enterprises need to have a good and clear view of how to use the technology to add value to their unique business. Comparing specific metrics of cloud computing and traditional IT can be helpful in evaluating costing and work efficiency. The cloud ROI model must include components like workload-wise assessment, capex versus open costs benefits etc.
- RELIABILITY AND AVAILABILITY
  Cloud providers still lack round-the-clock service; this results in frequent outages. It is important to monitor the service being provided using internal or third-party tools. It is vital to have plans to supervise usage, SLAs, performance, robustness, and business dependency of these services. Cloud computing often suffers from frequent outages, owing to the lack of round-the-clock service on the part of cloud providers. It is important to monitor the cloud service on the part of cloud providers. It is important to monitor the cloud service continuously as well as to supervise its performance, business dependency and robustness.
- PERFORMANCE AND BANDWIDTH COST
  Businesses can save money on hardware but they have to speed more for the bandwidth. This can be a low cost for smaller applications but can be significantly high for the data-intensive applications. Delivering intensive complex data over the network requires

sufficient bandwidth. Because of this, many businesses are waiting for a reduced cost before switching to the cloud.

- DATA MANAGEMENT CHALLENGES
  The main challenge is to provide ease of programming, consistency, scalability and elasticity at the same time, over cloud data. It has sacrificed consistency and ease of programming for the sake scalability. Force applications to access data partitions individually, with a loss of consistency guarantee across data partitions. While the developers faced with a very difficult problem, providing isolation and atomicity across data partitions through careful engineering. All these challenges should not be considered as road blocks in the pursuit of cloud computing. It is rather important to give serious consideration to these issues and the possible ways out before adopting the technology.

Cloud security is one of the most important and complex task as it leverage the privacy and resources of individuals and enterprises. Security of cloud is a challenging task because it deals with different levels of cloud whether it is private, public or hybrid cloud and also attack can be external or internal. The focus of the paper is to identify and discuss the main security attacks on cloud. Rest of the paper is organized as follow. In section 2, a brief introduction of security with challenges is provided. In section 3 and 4 the attacks has been explained with possible solutions. Finally the paper is concluded with future work in section 5.

2. Cloud Security

Attacks on the cloud can be external and internal. These threats [7,8] are very much similar to attacks faced by various data center vendors. In this security responsibility is divided between the all the shareholders like cloud users, cloud vendors and the third party vendor to provide secure software and configurations. For providing the security at physical layer firewall can be installed. Besides all these external security attacks, cloud does face other internal security challenges as well. Cloud host must provide the protection from theft or denial-of-service attacks by users, which mean the malicious behavior of different users, need to be checked. Virtualization is one of the techniques these days clouds are using to protect from other illegitimate and malicious users. Though, some of the virtualization techniques suffer from error. Sometimes incorrect virtualization allows access users to access sensitive portions of the provider’s infrastructure or to the resources of others. The cloud should also be protected from the host because it directly has the access to bottom layer of software. Therefore the security aspect needs to be dealt at every possible angle.

3. Attacks

There are various types of cloud security[8] attacks caused during the growth of any business strategies. They are:

Wrapping Attack Problem- Whenever user makes a request through their browser using virtual machine, it will be directed to respected web server. The attacker encloses the message having the structural information that will be exchanged between the browser and server during the message passing into a new header, the <Wrapper> element. In this, when a user makes request from his VM through the browser, the request is first directed to the web server such that SOAP message is generated. In this, if an attacker changes the structure of message and one of its properties is modified, then the attack can be detected. This approach is known as scheme validation.

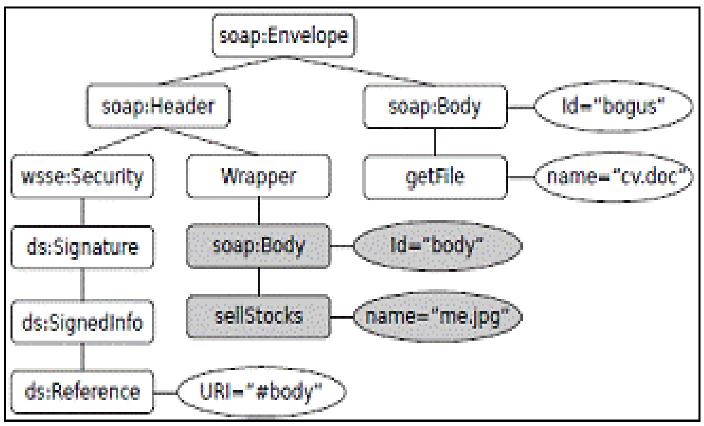

Figure 1 Soap Wrap Message

Malware-injection attack problem-In this attack, the attacker causes problems during the exchange of meta data or makes his own instance or advisory to intrude with malicious service so that it may look like a valid instance service running in a cloud. In this, the client request is based on authentication and authorization such that there is a huge possibility of metadata exchange between the web browser and web server.

Flooding Attack Problem-In this attack, when an adversary has achieved the authorization to make a request to the cloud, then anyone can easily create fake data and requests to the cloud server. When processing is done, non-legitimate requests must be checked to determine the authenticity, but checking CPU utilization is more, memory storage is also high and IaaS engages to a great extent, as a result the server will offload its services to another server, thus interrupting usual processing of one server.

Accountability check problem-In this attack, attacker engages the cloud with a malicious service or code, such that it consumes a lot of computational power and storage from the cloud server due to which, a dispute arises and the provider's business reputation is hampered.

4. Possible Attack Solution :-

Wrapping Attack Problem Solution-To solve this problem, the registered users of cloud computing can approach two types of precautions. Firstly, Registering a public certificate with the provider and Secondly, by generating a self signed certificate and RSA key for convenience. All these certificates must be authenticated by a trusted CA. So that if security is breached by chance, then actions can be taken immediately and accordingly so that SOAP message can be ignored.

Malware Injection Problem Solution-To solve this attack, hypervisor is used as it is responsible for scheduling all the instances from the FAT –like table of customers Virtual Machines(VM).This solution helps to utilize a FAT-like table(File Allocation Table)System architecture due to its straightforward technique which is supported by virtually by all existing operating systems.

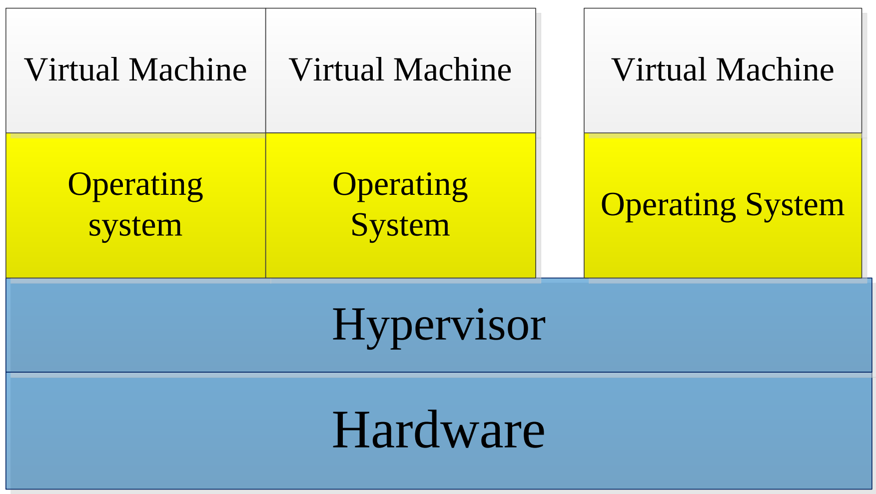

Figure 2 : Guest OS identification

Flooding Attack Problem Solution-To control this attack, we need to consider all servers in cloud system as a fleet of servers, such that each server is designated for a specific types of job. In this all the servers present in the fleet will have internal communication among themselves through message passing, such that whenever a server is overloaded, a new server will be deployed in the fleet and the name server, which has the complete records of the current states of the servers, will update the destination for the requests with the newly included server.

Accountability check problem solution-In this attack, our main purpose is to provide some features in cloud computing so that the customers don’t face problems during the account check. They are:-1.Identities, 2.Secure roads, 3.Auditing, 4. Evidence. Firstly, before starting instance, the identity of legitimate customer should be checked, this is done by hypervisor. Secondly, all the messages passing and data transfer in the network will be stored securely and uninterrupted in that specific node. Hence, when the auditing takes place, all necessary information can be retrieved. Using this approach we will keep the privacy of both the providers as well as customers, by keeping the log of all the records available to only the victim customers and not to all the customers in the cloud.

CONCLUSION

Cloud computing is benefitting the Information technology industries and resources in their usage and management but new technology has lot of challenges. In the paper we have mentioned some of them and proposed possible solution for attack such as using FAT-like table and a Hypervisor. In our future research work, we will further extend our research by implementing and analyzing the results using the simulation tools. Well-developed security architecture will give customer confidence in the field of cloud computation.

Acknowledgement

We would like to express our sincere gratitude to Accendere Knowledge Management Pvt. Ltd. for providing us the platform and opportunity to pursue the research.

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